

Prof. Karl Friston ~ Applied Active Inference

Symposium pt. 2 (Communication)

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hello welcome back this is the second session of the active lab symposium on june 21st 2021 and we're here in the dot-coms or communication organizational unit the goal of dot coms is to organize the labs internal projects and activities as well as to carry out all forms of communication with external entities so it's like our connective tissue and our neuroectoderm in a way what has been done so far is that we've had about 75 live streams variously on presentations and participatory discussions since july 28 2020 and we're taking an active inference approach to communication and learning by doing our very first live stream was on the paper narrative as active inference and shortly after a world unto itself human as active inference so that's something that we like thinking about here and that we want to explore and also some of our lab members framed online communication and team collaboration in terms of active inference in the 2020 paper active inference and behavior engineering for teams our aim here is to make active inference accessible well-known and well understood so let's get right to the questions carl the first question is how can we show not tell the idea of active inference for example through embodied experience experiments or what other mechanisms how can we best communicate in a way that makes it resonate um well pursuing that very interesting notion of using the principles of active inference to optimize the role of the dot coms team um then at its simplest active inference means that the imperatives for all behavior and it's likely that most behavior is of an epistemic sort is to resolve uncertainty so if you want to engage people um and be a um a service um to the people you want to engage which may be um internal members of of your own team and then you've got to know um what they don't know um because that will define the um the epistemic affordances that will um get them engaged with you and you with them um that will um incur the best kind of belief updating so um you know practically what that might mean is that it may well be that you um one has to identify um didactic or informative illustrations um that are tailored or specialized to the person or people that you're talking about um so um and then you know i have never thought

about using embodied experiences before but that's a brilliant idea you're [Music] just illustrating to people who want to know how my body and my mind works to you know illustrate to them you know the mechanics of it working using the language of active inference it can be very very powerful so as i'm talking you know what one example of this would be um using psychotic suppression as an illustration of the potency of getting your beliefs about the predictability of sensory evidence right um so if you're a physiologist um you would you would know this as sensory attenuation if you were in machine learning uh you're working with transformers this would be i think attentional selection um basically deploying the gain or switching on the right channels in order to select those data that are going to resolve the most uncertainty or maximize the uh the information gain um subsequent on this sort of covert action or you know often sold as a sort of mental action um in the philosophy literature so that mental action um is is really endemic and a vital part of our sensory engagement with the world and beautifully illustrated by sarcanic suppression so this is the um speaks to the notion of um attenuating the sensory consequences of your own action so that any evidence that you're not actually acting is precluded from your belief updating so a clinical example of this would be parkinson's disease for example if i'm sitting still and i wasn't able to ignore all the messages from my muscles that tell me at the moment i am sitting still then i am never going to be able to realize in a priori intention that i'm going to initiate a movement because as soon as i initiate a movement i have i put in place a plausible hypothesis that i'm lifting i'm going to stand up um immediately all the evidence at hand suggests that i am not standing up so i'm going to revise my belief no i'm not standing up i'm not in the process of standing up and it becomes impossible to move um so that would be an example of what would happen if you didn't have this capacity to to or to select or apply the principles of optimal bayesian design in terms of selecting those data for your own belief updating but a really pragmatic and easy example of that is psychotic suppression when we do the simplest of movements um epistemic foraging which is moving our eyes making psychedelic eye movements because when we do that we actually induce masses of visual information on the photoreceptors in the retina uh sometimes referred to as retinal slip so when i look from the left to the right there's a flood of information that i have caused and yeah it is not useful information because it doesn't tell me anything i didn't already know so what the what the brain does it um it it suppresses that information by transiently suspending the precision if you you're filter like perspective on um predictive coding as what kind of um variational filtering um and that's really easy to demonstrate to an audience you know just get them to to either fixate on essential uh stimulus

um and then pay attention to something that's moving around or the converse they fixate on the thing that's moving around and then ask them questions about the central stimulus and with the right timing it's a very potent illustration of um beyond just gathering information but actively um selecting and triaging that information in accord with the principles of your optimal optimal days in design so i haven't thought about going beyond that but i'm sure there are lots of lovely um examples um of embodied experiences that really do illustrate um the um the you know active inference in action as it were i'm just reminded um because active inference could be read as if you like a um a 21st century version of ideomotive theories which were very popular in in the 19th century and and of course that was um demonstrated through embodied experiences in a very alluring way through hypnotism and the like so i can i can imagine suddenly doing a sort of uh 21st century version of hypnotism and all those wonderful victorian illusions about um you know the way you use your sense organs or deploy them actively and but now in the service of just illustrating some basic phenomena that underwrite active inference in terms of experiments or you know the classic um ones that immediately come to mind that are really engaging um are visual illusions so on one reading um all visual illusions are just ways of getting out your perceptual priors in the context of of bayesian inference that you know if you can conjure a particular pattern of sensory information that you know was caused in one way and yet you think your subject or your audience has sufficiently precise prior beliefs that it could only be caused in another way which is not the way you caused it and then you let them experience that and then you reveal how you actually generated those data then that's a very powerful way of demonstrating the innate priors the sort of formal prize in you know in terms of the connect term or the um the the sparse coupling on a on a factor graph and the again is part of the lived experience so i think visual illusions would be and there are loads of beautiful illusions out there and all that one would have to do is to harness their beauty and the law and use them as a vehicle to give people insight into the their own usually sub-personal prize about the way that the world the world is constructed and then um i would i mean in my world what you generally try and do is to actually put this in silico by just you know creating little silicone creatures and because you've now got active imprints with this information geometry and you you now have the opportunity not just to simulate what these creatures do but what they perceive because now you've got the um the quantitative estimates of their their posterior beliefs so you can actually show a subject who's just experienced a visual illusion that this is perfectly base optimal and indeed when you write down this variational message passing scheme in this synthetic subject

uh this synthetic person also experiences exactly the same illusions and this is and this is and this is per basically for this kind of kind of work so you can leverage activities in that sense and deliberately referring back to the edu discussions you know what active inference brings to the table because it's got information about stuff out there in the numerix you can go a lot further um than you can if you were doing say deep learning or machine learning because you've got this um this dual you've got this information geometry at hand your the state of your variational autoencoder actually means something in relation to a belief about what generated the those data which you can create lovely little movies you know showing what this um simulation of you um was actually experiencing so um you know i'm interested are there are there any other ways that you you've you've thought about in terms of showing people thanks for the answer it's like look left look right now you're an active inference agent and as far as potential avenues for embodiment some of the work with ryan smith and others bringing people into the somatosensory dimensions and their own priors and expectations about their body and about motion could be very powerful as well as auditory modalities and indeed active inference is a framework by which we can think about how our perceptions are related to our inference and our action so in various domains i think they'll be excellent experiments and it brings us to our next question you actually addressed several areas in your answer you addressed machine learning as well as neuroscience as well as just everyday lived experience so how can active inference engage in better dialogue with adjacent areas for example machine learning systems psychiatry and neuroscience as well as any other fields that you think are relevant to um then you know the obvious answer here is in either academic or commercial um and what might you what would license that um i think um you know the simplest answer is that the free energy principle and it's if you like technological coral reactive and is not there and was never intended to um replace uh extant theories it was there to prove uh to endorse them and to reveal the interrelationships between them so anything that's worked and survived into the 21st century has some veracity and a proven utility and therefore it's just a question of um reformulating or changing the words so that people can see immediately um how their particular formulation relates to somebody else's formulation where both formulations are special cases of the most generic and simplest explanation which would you know from my point of view would be the uh the free energy principle and active inference in the case of uh sentience so i think that's you know as an a an integrative framework you're very well positioned to say look you know can we understand the way that you think about this and um can we now um articulate this um either using simulations or you

know mathematical analysis um okay you know can we understand what you've been doing in this integrative framework and if we can um can we show how it relates to another disciplines formulation of this problem and sometimes you can get synergistic or added value from doing that so you know there must be loads of examples you've written machine learning systems engineering psychiatry neuroscience here so machine learning for example you know how would active inference help machine learning um so at the moment um you know there seems to be two um answers um floating around um and we've we've already sort of discussed a couple of the uh a couple of the key a couple of these issues in depth um so machine learning commits to usually a normative approach to good behavior that can be quantified by a loss or a value function um but we've just said well if we now want these machines to learn to act then we have to go beyond um state action value functions and consider the belief based calculus that is active inference which is all about the reduction of uncertainty so now you are in a position to say well look if you consider your objective functions as a part of a more generic objective function um do you think what you might be able to get from this and of course what you might get from this is a deep learning scheme that actually can now go and solicit the right kind of data to optimize its own learning and you know people in bayesian rl might argue well that yeah that's what we're doing uh you know with a series of bright ideas and and heuristics to try and augment um classical value functions but you know you can say well okay you probably put a lot of work into that but there is there is actually um you know a simple objective function already out there that is provably um uh to describe um systems that self-organize and maintain themselves that actually has what you want why don't you try this for example um so that would be one example you have to tread carefully because you know a lot of people have dedicated their lives to solving these problems and they're very reluctant to change their rhetoric or or see their contributions as a special case but in many instances um certainly from my perspective mathematically they are special cases and sometimes if you catch the um the the entrepreneurs the innovators the creative academics at the right stage in their career before they have committed to a particular church or ideology or calculus or group or company and you can you can actually point them in the right direction they become extremely creative and i'm not so sure about systems engineering but certainly i always celebrate um the um the expected free energy as with just taking away various bits and pieces very sources of uncertainty as um reducing to um kl control and then what i say is what chaos control is what grown-up engineers use in the control theoretic setting and so that would be another example so um you know you could also i don't know this because it's not my field but

certainly in in in terms of introducing say a fault tolerance in um in control theoretic approaches in where the fault tolerance required uncertainty about the operation of some external part you could again motivate a more complete objective function that takes you beyond kale control and introduces the information gain in your in in into the mix because to get from the complete objective function or to kl control you have to ignore uncertainty about the latent states that are in the mapping from latent states of the plant you're controlling to the the the sensors or the observables and so you you know you're moving from a partial observed markov decision process for example to an observable one and then um accepted the energy becomes kl control or risk sensitive control in economics so you could you could say well look you know why did you just augment the the your kl control and then put this extra term in and now what you've got is is a kind of anticipatory fault tolerance in the sense that if if there's uncertainty about latent causes that's automatically resolved in the way that you um go and um switch on various sensors or switch off various sensors as you know there's a principal way of doing that i think to have any influence you need to be able to show or provide proof of principle that this um more integrated more um universal um approach to normative normative approach to problems um can offer um speed ups or increased efficiency or do what the people actually in that field wanted to do so for example you've got to be able to show that active inference can outperform um sort of vanilla deep learning by an order of magnitude which is easy to do because of course most most benchmarks and machine learning are actually inference problems so if you just recast it it's actually quite quite a trivial thing to do just by saying well actually what you've been dealing with is an inference problem uh which looks a lot like a one-shot learning from the point of view of somebody um in machine learning but i think there will be some pressure to get people's attention um to make yourself um attractive in terms of you they will now have some first of all um find you um um interesting um and also uh have the potential that they can place an epistemic trust in you you've got to you've got to sort of you know give them a cue a clue and a cue as to why they should uh should engage with you and and very often there's that there's a two-way uh two way or two-road exchange um so one simple example that which i see emerging in the field is the use of um um you know so deep learning to amortize certain mappings um when they can be amortized in active infant schemes to evaluate you know the expected energy for example or doing very deep tree searches so that's that's the kind of innovation you've seen coming out of 20 year olds at the moment who haven't yet decided whether they're going to do deep learning or active inference because they want to do both and do it very very effectively so

that that's a nice example from my perspective on the sort of you know that integrative role that could be played or you could play thanks i i really heard this yes and maxim from communication and improvisation it's like yes there's been a disciplinary way of approaching it and we're going to be working together to come back to first principles or to make it more efficient so that's really powerful how does active inference help us rethink the nature of online communication where so much of our communication nowadays does occur that's that's that's a big question isn't it um and certainly in the context of um social media politics fake news and the like you could you could take that question in lots of lots of different directions which i won't do because that's not my field of expertise um but just yeah off the cuff in terms of first principles and what is um it's the ability for me to infer what you meant it's the hermeneutics problem um if it's a hermeneutics problem that's most efficiently um resolved in terms of um dyadic or multi-system interactions um when we um come back to first principles which is the genetic model when we share the same narrative or same generative model um so in terms of helping um actually how does it um how does active influence help us rethink the nature of online communication and i think just from a first principles point of view it would be the importance of establishing who is talking to who and if you want to optimize the efficiency of that exchange literally from the point of view this principle of least action the speed with which you can resolve uncertainty and minimize your uncertainty or surprisal and then it's ensuring that like-minded communicators are actually communicating because it's only them that will understand each other so everybody has to speak the same language they have to commit to a shared narrative and a shared genetic model and then just by things like rate distortion theorem or you know rewriting that in terms of uh 300 min the joint free energy minimization between two interlocutors and that's the most efficient sort of shared path of least action um how does that help engineer or um intervene on things i i'm not so sure um certainly just in reference to communications with uh you know people um people like maxwell and the like and other colleagues um there is this interesting notion that if communication if the real problem of communication is not really the messages that you send but the inferring whether to send the messages to this person or not that itself now becomes conditional upon inferring that's a member of my in-group or that's a creature or a person like me and then the question is um you know how do how does self-organization say you know in terms of social media exchange how is that underwritten by an inference about the kind of people who i am listening to or who i am talking to and what are the basic principles of that and again in accord with um the minimization of complexity in our genitive models what we

you know what then it may be a useful hypothesis to say that there's an inevitable course graining of the way that we conceive of the people that we generate information for say on social media and reciprocally and the kinds of people that i will be able to solicit by listening to this twitter feed or that wikipedia page or this news channel so understanding how people carve up whether they are like to the degree of similarity to them may be very useful in just getting an idea of the dynamics of message passing amongst amongst communities that will be defined by on average how each member of that ensemble or individual course grains and has a genetic model of the kinds of people in the communication crane and just to finish this which is something i've heard and i found it really interesting notion um that again would be great if one could simulate this and understand the maths behind it is that the only evolutionary stable um from the point of view of the free energy principle the only one that will be selected by a processor bayesian model selection the only the only partitioning into in groups and out groups is a 50 50 in group out group and in the sense that anything that departs from that sort of dynamically unstable but evolutionary stable partition means that the smaller group the outgroup the the the old man out will necessarily ultimately be absorbed into the into into the um into the larger group so the only the only stable um um uh partitioning is 50 50 which which makes a lot of sense when you look at trump versus biden when you look at brexit versus not brexit wherever you look all the important allegiances in terms of our political ideological uh possibly even theological communication seems to be split right down the middle and perhaps it can be no other way so it'd be very very interesting to simulate that and see if that is a truism that inherits from um all of these um marginal likelihood or free energy minimizing processes implemented at multiple levels of hierarchical um multiple hierarchical levels which is you know communication is just message passing and message passing is just the way you articulate updating and believe updating just is the process of inference which just is the the paths of least action according to the fvp the 50-50 politics it's maximally confusing something we all experience and a few key points there about the nature of online communication is that at the core it is dyadic even when you're broadcasting to many it's actually about that connection and the hermeneutic relationship of unpacking meaning and then also you brought up the importance of context and identity and who's talking to who and our inferences about that which is essential to rhetoric and something that often gets left off when people take big data approaches to online discourse the next question is how does active inference help us think about science communication and participation specifically as we move into broader citizen science initiatives and scientists are in the loop something you've

been recently involved in as well with society and with decision making so as science and the nature of science is changing who is doing it and how they communicate it how does active inference help us navigate that right well i'm sure you've thought about this much more deeply than i have it's um just drawing upon my experience um you know in terms of science communication uh during the um the coronavirus epidemic yeah i think you're absolutely right you can as with the previous questions i think you can take the principles of um and just think about um what does that mean for optimal communication and um and belief updating and shared belief updating and shared narratives or a lot and um use that um as a point of reference for the way that you articulate your own science and you've asked all the your challenging and exactly right questions you know about how you communicate how you engage other scientists or other um other partners within or beyond academia and i think the same principles apply exactly to um the the public and just to reinforce your i think beautiful observation that all communication is dynamic from the point of view of the person communicating you know so it's this kind of person as a unitary object i am talking to or this population or this mentality or you know so on this discipline so it is i think fundamentally you know dyadic from the point of view the person generating the messages um that will uh may or may not incur belief updating in the recipients and these kinds of principles i'm sure would be useful in terms of science so at that level um i don't know that there's much that i would have to what you already know and you know and will all and possibly already are implementing um there is another level though which is um using active inference not as a model for the way that we work and communicate and participate but as a um as a more a a a statistical or observation model of data so um in a sense um you can use the principles of active inference really to make the most of data pertinent to a particular domain so again i'm thinking here of the dynamic causal modeling of the epidemiological and behavioral that has been generated by the coronavirus epidemic um so um you you you can you certainly use the um the perceptual inference side of it if you like the bayesian filtering side of it and but also in principle the um the data mining or the you know the the the optimal bayesian design to select which data are useful or not in a very practical way when assimilating big data in the service of understanding the um the system at hand so the system at hand is how does a um how does a um a spike propagate from one neuron to another neuron a neural network or how does a virus propagate from one person to another person in a neural population or a network a population network and then you can certainly use the data to um you can apply active inference to build genetic models of how you think that occurs and what immediately comes confronts you is you've got

to put in all of the things that generate those so you can't miss out any factors um that are important to be the psychological be the behavioral um be they viral be they um transport related all of these things have to go into your generative model to best explain the data so you know when we do this in a practical way we we both use um the instance rates from per testing and google mobility data and department of transport transportation data anything that speaks to them and reduces uncertainty about all the factors necessary um they're entailed by your generative model in this instance not a discrete space it's written down oh no this one is actually a discrete state space model um and the active influence is not explicitly part of it in the sense that we're not trying to predict people's behavior um but it it does serve um an indirect um guide through the principles of bayesian optimum optimum design and all that basically means is do i invest computational resources and thereby incur computational and statistical complexity by including or attending to this kind of data or not and then you can actually evaluate the the the information gain by including that data of that data so for example you need google retail estimates and and or workplace activity or just one uh if you include both that means that your the complexity increases and your uh literally you have to wait another half hour before before you get the results for your dashboard and or do you or do you not and have a more parsimonious um have a more parsimonious model in the sense that you have now in the same sense of that circadian suppression of retinal slip and you've actually said no i don't need that i've got everything i need i got the right kind of data just by focusing on these data and then once you've got that in mind you can now go foraging for different kinds of data different collaborators from different who've got different perspectives but also crucially different data you know to try that will inform and shrink your uncertainty about the model parameters and also very importantly about the structural form of the model do i need this node how many um is this interaction important or not is this degree of nonlinearity justified by the data so all of these questions affected your hypotheses about how this system is responding or would respond if you intervene on it all of those questions now become meaningful to an evidence-based analysis because you've got a generative model underneath the hood so that would be a more practical application of um active the principles that underwrite active inference even though your computer program is not actually doing active inference but it's certainly been deployed using the principles of active inference awesome and we heard that integrative approach yes we're going to include multiple data sets potentially of unconventional type and we're going to have a principled way of deciding how to include that data and also as you brought up at the end who to include in the conversation and there was

one piece you said in there about the dyadic nature of communication where a speaker is always i think you said something like speaking to a person or to a group or to a community and it relates to our next question which is how can we appropriately interact with shared and nested generative models potentially across scales be it person team or community do we think about these levels of analysis as active inference agents in their own right or how do we for example speak to a community or speak to a level of analysis that's broader than the personal yeah i think that's a great and challenging question um i mean clearly there has been some provisional work in academia uh looking at sort of um mark of blankets of markov blanket which is effectively from a stats point of view from a physics point of view at least what we're talking about here as a physicist um you'll be tackling this with things like the renova or the apparatus of the renormalization group which tells you immediately something um interesting that the existence of um this nested structure um if underwritten by or if it is a renormalization group means that there are certain functional forms that are conserved so what that means is from your practical point of view that there will be certain kinds of behavior um that are actually conserved at different scales so what works in terms of talking to your children should also work as a president talking to your or a a governor talking to your state or a team leader talking to your on your your your assembled team um simply because um in order for there to be a hierarchical nesting that supports that hierarchical structure that has to be this conservation usually mathematically written down as a the functional form of the lagrangian um or it could be the sort of martial likelihood or the surprise that we're talking about um that underwrites these sort of um paths most likely paths or paths of least action so that actually paradoxically slightly makes the problem slightly simpler because what you're saying is what works at one level will work at all levels all you've got to do is um find the course grading operator operator that takes you from one level to the next um so what that would look like i think would be very very application domain specific so you know i think that there is a great challenge ahead which is taking um the single particle fep approach now to into a world that matter where it matters where the world is actually an ensemble of particles and we've already discussed the importance of um you know thinking about worlds where all the particles are identical whereas all the particles actually vote for half the particles vote for trump and the other half vote for biden and this is interesting to reflect upon um pre-21st century physics that was so powerful in um articulating this kind of dynamics because it just dealt with the simplifying assumption that my idealized gas was an ensemble of identical particles and then you can spin up from that equilibrium physics and everything that led from the um you know

from from carna cycles and and and engines through to uh to to to current current technology so it's a powerful assumption if you just make some simplifying assumptions but just because you know we've already said well perhaps that's not the best kind of assumption to make when you're dealing with political uh mechanics at least like one bi partition in there and so that would require a revisiting of the of that kind of um physics um from our point of view or your point of view basically simulating uh active age inference agents um or ensembles of active inference agents particles but where now there's a heterogeneity in play and then asking the questions well what are at the next scale um the free energy minimizing or surprising minimizing or potential minimizing um solutions at the next scale up so we come back to our you know whys in the case of people are all split 50 50 which has an enormous impact on the interactions at the scale below so i think to tease i think all i'm just hand waving here because i don't think there are any um any formal answers and i think those formal answers will probably have to come out of agent-based and possibly stochastic agent-based um modeling initiatives but with the twist you're making each agent itself an active inference agent so while each individual member of the ensemble is trying to minimize their free energy also the ensemble through cooperation and the shared narrative is minimizing the joint free energy and what that means for um when you move from one scale to the next scale i you know this is if you if you're in physics i imagine that this is the problem of beyond non-equilibrium steady states um because you're actually now dealing with the multi-scale aspect of non-equilibrium so at best we have good models of turbulent flow and b center your actions and laser physics that take us beyond equilibrium physics where all the particles are the same um into non-equilibrium physics but i don't know that there's an equivalent maths or metaphor in um in physics that would really speak to the the hierarchical nesting so i think this is a really open and important research area that i can only recommend is dealt with by numerical analyses basically predicated on underlying principles thanks for the answer there and it made me wonder if agree to disagree is a narrative that can be shared even when there is a 50 50 split and i think it brings us nicely to the final question of dot-coms which is how can we move people and teams into a co-transformative space or as some of recent work discussed in interactionist space well i'm sure you know the answer to some of these i don't know i suspect now i'm now realizing you already know the answers because you're knowing smiles when i say something that you recognize but um so actually answering that from on the basis of what you just um what you just said um so yeah i think that's another really uh useful insight that you know agreeing to disagree is a a surprise minimizing bayes optimal um explanation for

the exchange with others and so but it does rest upon committing to the hypothesis that you are not like me you are not like-minded and that's okay so i've now classified you as somebody who's not like-minded and i've resolved the ambiguity in among the hypotheses that you are either like-minded or you're not like-minded normally we resolve those within a few your first impressions within a few seconds based upon all these epistemic cues we offer each other to define the kind of you know sort of person that we are so we make that job as easy as possible for us and signaling uh to make this so we know our place and i i use that phrase because of course there's a paper called knowing your place that exploited um a shared genitive model that allowed you to be in a particular position in some space even if you should if you and i share the same understanding of political ideology but i know my place because i'm right-wing and you your place because you're left-wing or vice versa and then we can quite happily exchange but disa agreeing to disagree so i think that that's that's a wonderful um know perspective to have and to endorse it and that is a base optional perspective from both sides of the disagreement that's resolving uncertainty and you know in a in a bounded rational way so applying that notion to co-transform to space it reminds me of the problem where certain patients in psychiatry have committed to a particular inference the you know whether they belong out there in that kind of environment or not so that and they have decided um that they do not belong out there and that those people are not like them um and they start to avoid so in a very simple-minded way and if they're any psychiatrist in this example but i think it's illustrative and useful in that respect so say take depressive depression or agrophobia you know which is a completely baze optimal response if i have committed to the that out there is full of people who are not like me and potentially will upset confuse and render me uncertain and possibly even injure me in some way so withdrawal um into that corner of uh into your house or into that silo if you're working in you know you know in teams is a perfectly based optimal response says that you've got your a precise belief that this is where you belong this is this you know this these are the people that you speak to and not those people so and that's usually perfectly functional um in psychiatry that would be a neurotic defense but it can become pathological um when you become house bound or you become say if you've got a pathological hypothesis like your body got dysmorphophobia and you you get you know you nearly die because of a failure to eat properly so when you say co-transformative i imagine what you mean is you want to transform two teams into uh one team or at least enable them to work together if that's right and you're nodding partially so assume that it is uh you're facing the same challenge um that a psychiatrist faces in terms of enabling people to revise the

precision of their precise beliefs about who they can interact with and who they should interact with that's not an easy thing but it's certainly doable and it basically usually reduces to um presenting evidence to a group or a person that it can be another way so that they start you start to um revise their prior beliefs or at least the precision of their prior beliefs by in a safe space where it's okay to explore other hypotheses um enabling them to think about other ways of interacting um so this would normally be the objective of psychotherapy basically um to by illustration very much in the same way you were talking about sort of you know illustrating or educating by embodied experience very much psychotherapy is thought to work like this you provide a psychologically embodied experience where you can try out different styles and different hypotheses and in so doing you paradoxically introduce the right kind of uncertainty that about different styles of engagement and who you are and who you are talking to and by relaxing that precision you enable um you give the patient or the or the naughty team that's become too siloed the latitude to explore other ways or other ways of behaving so i would imagine that most of um most of the tried and trusted in procedures um to get teams into a co into a co-transformative space and you know use one of or more of those mechanisms what would active inference be bring to the table it would just bring the the narrative that all everybody you're trying to get to talk to each other um can come to share so they can see through the process of becoming um more collaborative or exchanging ideas more fluently or working to working with the cert with the same lexicon or mechanics or code and it will enable them it will enable them to um to having the same narrative will actually shape their prior space um and understand the mechanics of actually enlarging the the hypothesis space in terms of inter interaction styles so that was an incredibly hand waving answer but it was in part informed by almost well my understanding of the question um from the point of view of the psychiatrist who wants to transform the way that a patient relates to to her to her world oh and drugs can help and i mean that literally those drugs um that are responsible neurobiologically for setting the precision if you can temporarily suspend the in order to reveal other latent a priory hypotheses in terms of the way that i am or the way that i interact and the way that i behave or the way that i perceive that can actually have long lasting effects and that on bringing those other hypotheses to the table in the moment in subsequent interactions so perhaps the most compelling example of this um which is trending at the moment is the use of psilocybin assistive therapy particularly in terminal care so you know if you know you're going to die of cancer in the next six months there are certain hypotheses that are brought to bear in terms of how i would expect to feel um and how i engage with the world

and how i engage with my loved ones as a dying person who is near death and the ultimate loss those are not necessarily the best or most functional hypotheses or ways of being there are other ways of dying gracefully and gloriously but to get at them sometimes having a you know a um a managed um challenge to your 5 ht 2a receptors via things like psilocybin and other related drugs just allows you to um suspend for a moment your very precise beliefs about the kind of thing i am and allows you to experience other ways of being and perceiving which can be very useful when it comes to just trying out other hypotheses and you know in your this is your cancer journey but you know one can also imagine similar scenarios when you get locked in to a particular way of um interacting with uh either within a team or to between teams in a larger organization so that would mean you have to go on a retreat and take lots of magic mushrooms never thought i would hear from you professor fristen but there we have it drugs for teams thank you for this excellent interval with dot-coms unit we're going to take another five-minute break and we'll return for the final session for dot tools thanks again everyone and we'll see you in five minutes